ASSIGNMENT / NETWORK DEVICES

Router vs Switch: Ports, LEDs and Setup

A clear, picture-based guide to the **router** and the **network switch**: what is on their panels, how they differ, and how to use them in a small gaming cafe.

The two devices at a glance

ROUTER
Connects **your network to other networks** (the internet). It gives out IP addresses, shares one internet line and usually provides Wi-Fi.

Think of it as the post office that sends mail between cities.

SWITCH
Connects **many devices inside one network**. It gives you more cable ports and sends data only to the device that needs it.

Think of it as the office clerk who hands each letter to the right desk.

Ports you will meet (easy words)

Port	What it is for	Found on
WAN / Internet	Plugs into the modem or ISP line. This is the "way out" to the internet.	Router
LAN	Plugs into your own devices: PC, TV, printer or a switch.	Router and switch
Uplink	A port used to link one switch to the router or to another switch.	Switch
Console	Used with a special cable to set up the device from a computer (for managed devices).	Managed router or switch
Power (DC / AC)	Takes the power adapter or power cord.	Both
USB, Reset, WPS	USB: storage or printer. Reset: restore factory settings. WPS: quick Wi-Fi connect button.	Mostly router

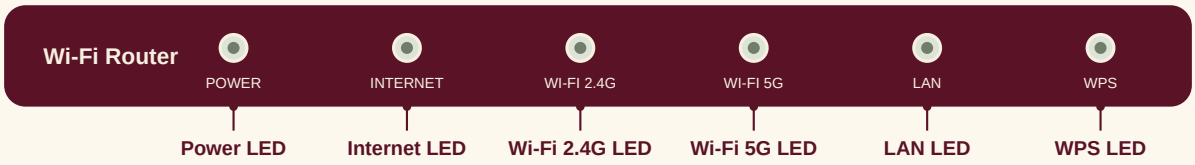
What the LED lights usually mean

LED state	Meaning
Solid / steady	The device or link is on and working.
Blinking	Data is moving (activity). Normal when you browse or play online.
Off	No power or no cable/device connected on that port.
Red or orange (some models)	A problem, for example no internet or a failed start-up. Colours differ between brands, so check the manual.

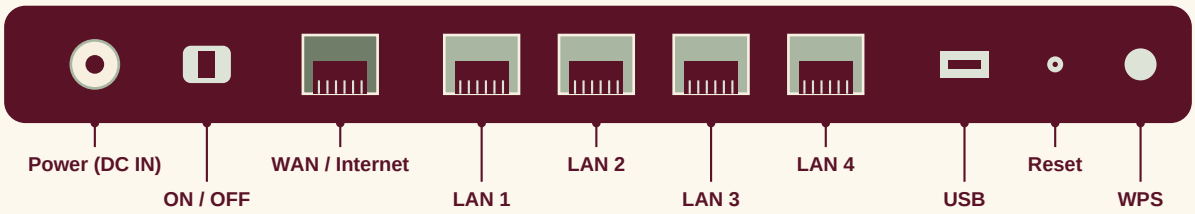
1 Labelled Diagram: Home Wi-Fi Router and Network Switch

Ports marked: Power, WAN, LAN, USB, Reset, WPS, Console, Uplink. Status LEDs shown on the front.

FRONT PANEL (status LEDs)

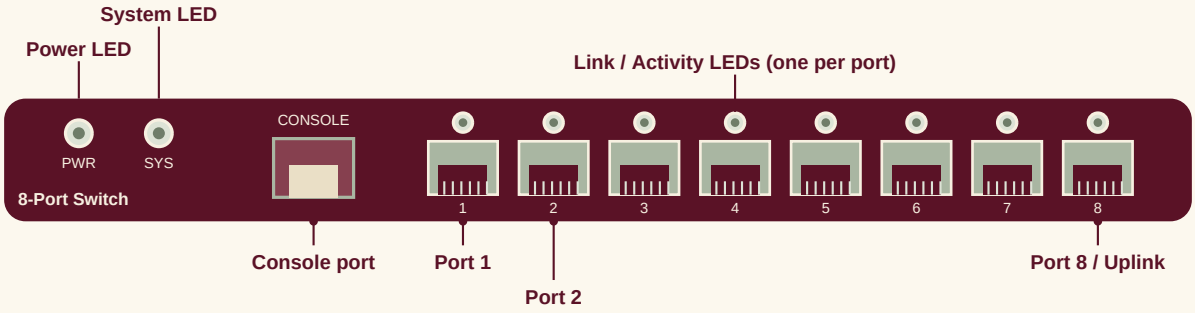


BACK PANEL (ports and buttons)



WAN is usually a different colour from the LAN ports. Antennas, vents and wall-mount holes are left out to keep it simple.

FRONT PANEL (ports and LEDs)



BACK PANEL (power)

Console port is found on managed switches only. Cheap home switches have just ports, LEDs and a power input.

Tip: Look at the front first for LEDs, then at the back for ports. On many routers the LEDs are on the front or top; on switches the LED sits right next to each port.

2 3 Key Differences: Router vs Switch

Main function, address used, and how traffic is handled

Difference	Router	Switch
1. Main function and network type	Joins **different networks**, for example your home LAN and the internet (WAN). It decides where traffic leaves your network.	Joins **devices inside the same network** (one LAN). It cannot connect you to the internet on its own.
2. Address it uses (IP vs MAC)	Works with **IP addresses** (Layer 3). It also hands out IP addresses to devices (DHCP) and can share one public IP using NAT.	Works with **MAC addresses** (Layer 2). It learns which MAC address is on which port and keeps a MAC table. It does not give out IPs.
3. How it handles traffic	Looks at the destination IP and **chooses the best path** (routing). It can filter traffic with a firewall and keeps networks separate.	Looks at the destination MAC and sends the data **only out of the correct port**. If the address is unknown, it floods it to all ports once to learn.

Simple analogy

A **router** is like a **post office**: it reads the city and street (IP address) and sends the parcel to the right network. A **switch** is like an **office clerk**: it already knows who sits at which desk (MAC address) and hands the parcel to that person only.

What happens when you open a website

1. Your PC sends the request to the **switch** (it checks the MAC address and passes it on).
2. The switch delivers it to the **router**, which is the way out of the LAN.
3. The router reads the **IP address** of the website and sends the request through the WAN to the ISP.
4. The reply comes back: router, then switch, then only the PC that asked.

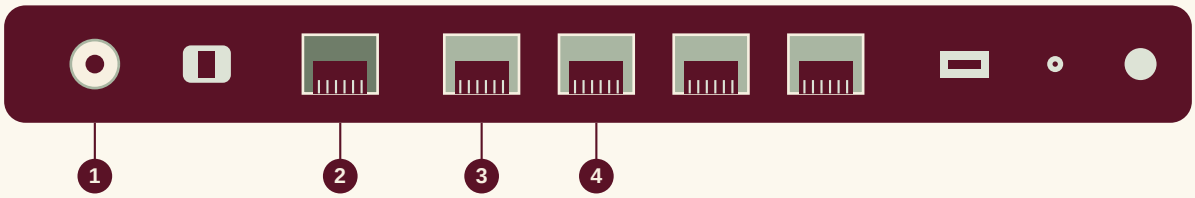
3 Annotated Images: Router and Switch

Power port, Ethernet ports and LED indicators marked with numbers

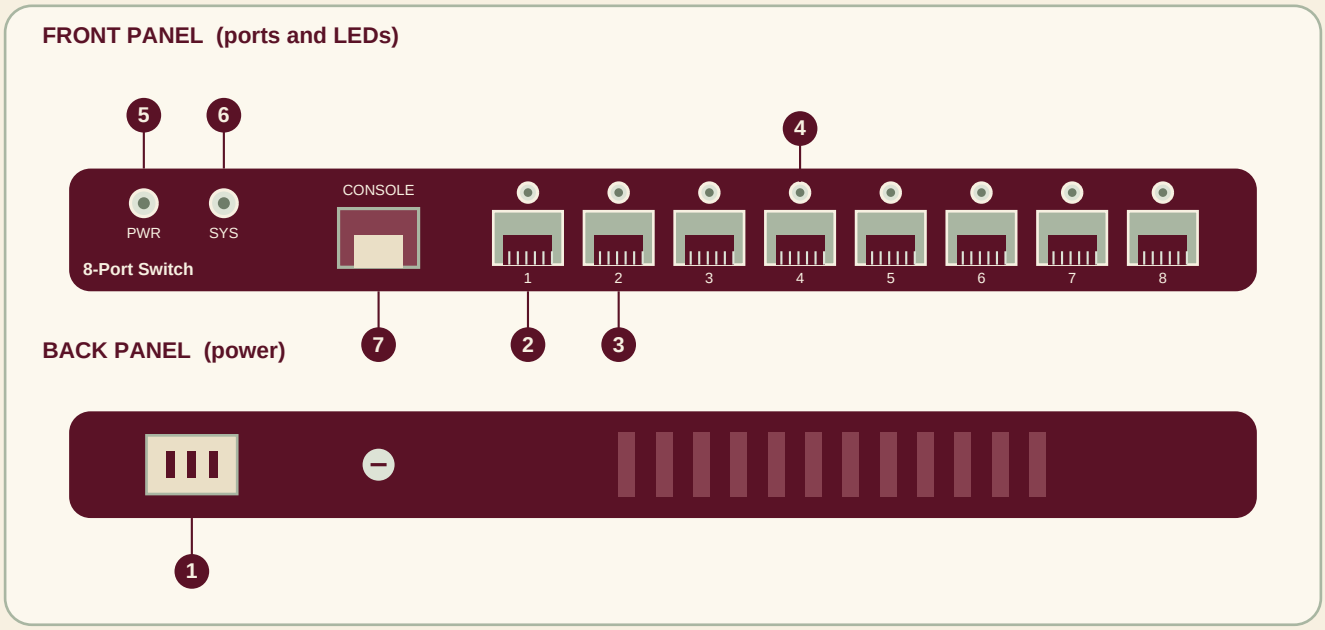
Virtual (drawn) images are used below. Numbers match the tables underneath.

FRONT PANEL (status LEDs)

BACK PANEL (ports and buttons)



No.	Part (router)	Position	Purpose
1	Power port (DC IN)	Back panel, far left	Takes the power adapter that switches the router on.
2	WAN / Internet port	Back panel, next to power	Ethernet port that connects to the modem or ISP line.
3, 4	LAN ports (Ethernet)	Back panel, middle group	Connect PCs, TV, printer or a switch. Many routers have 4.
5	Power LED	Front, far left	Steady = router is on.
6	Internet LED	Front, second	Steady = internet is connected. Off or red = no internet.
7	Wi-Fi LED	Front, third	On or blinking = Wi-Fi is working. Separate LEDs for 2.4G and 5G.
8	LAN LED	Front, fifth	On = a cable device is connected. Blinking = data moving.



No.	Part (switch)	Position	Purpose
1	Power socket (AC)	Back panel, left	Takes the power cord or adapter. Some small switches use a DC jack.
2, 3	Ethernet ports (1 and 2)	Front panel, row of 8	Connect PCs, consoles or other devices with RJ45 cables.
4	Link / Activity LED	Just above each port	Steady = cable and device connected. Blinking = data moving.
5	Power LED	Front, far left	Steady = switch is powered on.
6	System (SYS) LED	Front, next to power	Steady or blinking = the switch has started up normally.
7	Console port	Front, left of ports	Managed switches only. Used to configure the switch from a PC.

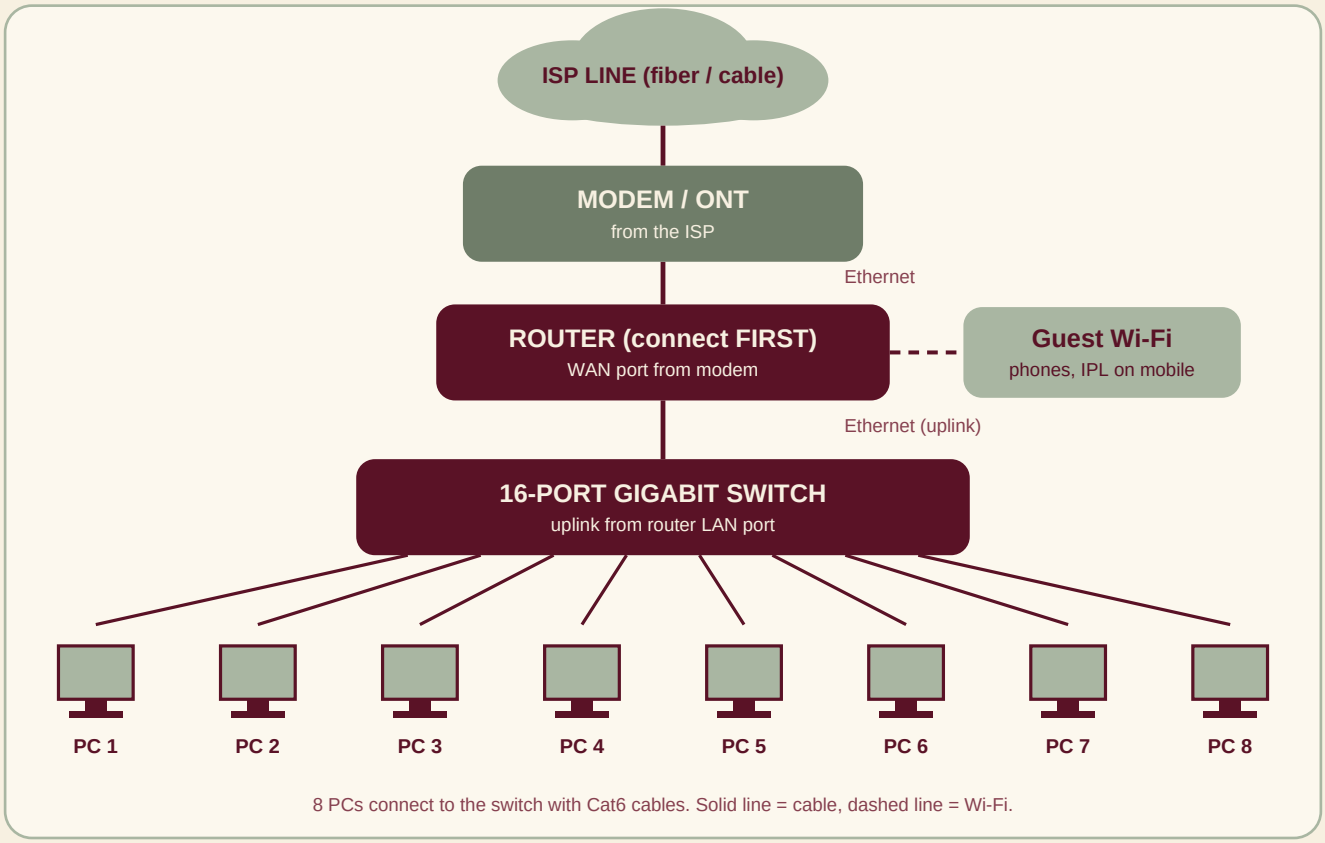
Using a real photo? Take a picture of your own router and switch (or save one from Google Images with its source link) and mark the same parts: power port, 2 Ethernet ports and the LEDs. Port positions on real devices differ by brand, but the purpose of each part is the same.

4 Scenario: Gaming Cafe for IPL Streaming and Multiplayer

Which device goes to the ISP first, and how 8 gaming PCs are connected

The plan in short

I would connect the **router first** to the ISP line (through the modem or ONT) using its **WAN port**. Then I would connect a **16-port Gigabit switch** to one **LAN port** of the router, and plug all **8 gaming PCs** into the switch with Cat6 cables. Visitors get Wi-Fi from the router.



8 PCs connect to the switch with Cat6 cables. Solid line = cable, dashed line = Wi-Fi.

Step-by-step setup

- **Modem/ONT to router:** plug the ISP modem into the router's **WAN** port. Power on and wait for the Internet LED to turn steady.
- **Router to switch:** connect one router **LAN** port to the switch's **uplink** (any port works on modern switches).
- **Switch to PCs:** connect PC 1 to PC 8 to switch ports 1 to 8 with Cat6 cables. Check each Link LED is on.
- **Set up the router:** keep DHCP on so every PC gets an IP automatically, and give each PC a fixed (reserved) IP for easy management.
- **Make streaming smooth:** turn on QoS to give gaming and IPL video priority, and use a separate guest Wi-Fi for visitors' phones.

4 Why these choices? (Justification)

Reasons for the router first, the switch second and the cabling

My choice	Reason
Router connected first to the ISP	Only a router can talk to the internet: it uses the WAN port, gives every PC an IP address (DHCP) and shares the single ISP line using NAT. A switch alone cannot do this, so PCs would have no internet.
Switch added after the router	A home router has only about 4 LAN ports, but I need 8 for PCs (plus 1 for the uplink). A switch adds ports cheaply and keeps traffic between PCs fast and local.
16-port, not 8-port, switch	8 PCs plus 1 uplink cable needs 9 ports, so an 8-port switch is not enough. 16 ports leaves room for more PCs, a printer or a streaming TV later.
Gigabit (1000 Mbps) ports	Gives smooth, low-lag gameplay and enough speed for HD IPL streaming on many PCs at once.
Wired Cat6 cables for PCs	Cables are more stable than Wi-Fi: lower ping, less lag and no drop-outs in multiplayer matches.
Guest Wi-Fi from the router	Visitors can use phones without touching the gaming PCs' wired network, which keeps matches fast.

Is the ISP speed enough?

Check the plan before you open. As a rough guide, one HD stream needs about 5 to 8 Mbps and online games need low *ping* more than raw speed. For 8 PCs plus guests, a fiber plan of **100 to 300 Mbps** is a sensible start. A small UPS (battery backup) also stops the network from restarting when power flickers.

Conclusion

Router first, switch second. The router connects the cafe to the internet and manages addresses, and the switch gives enough fast wired ports for all 8 gaming PCs.